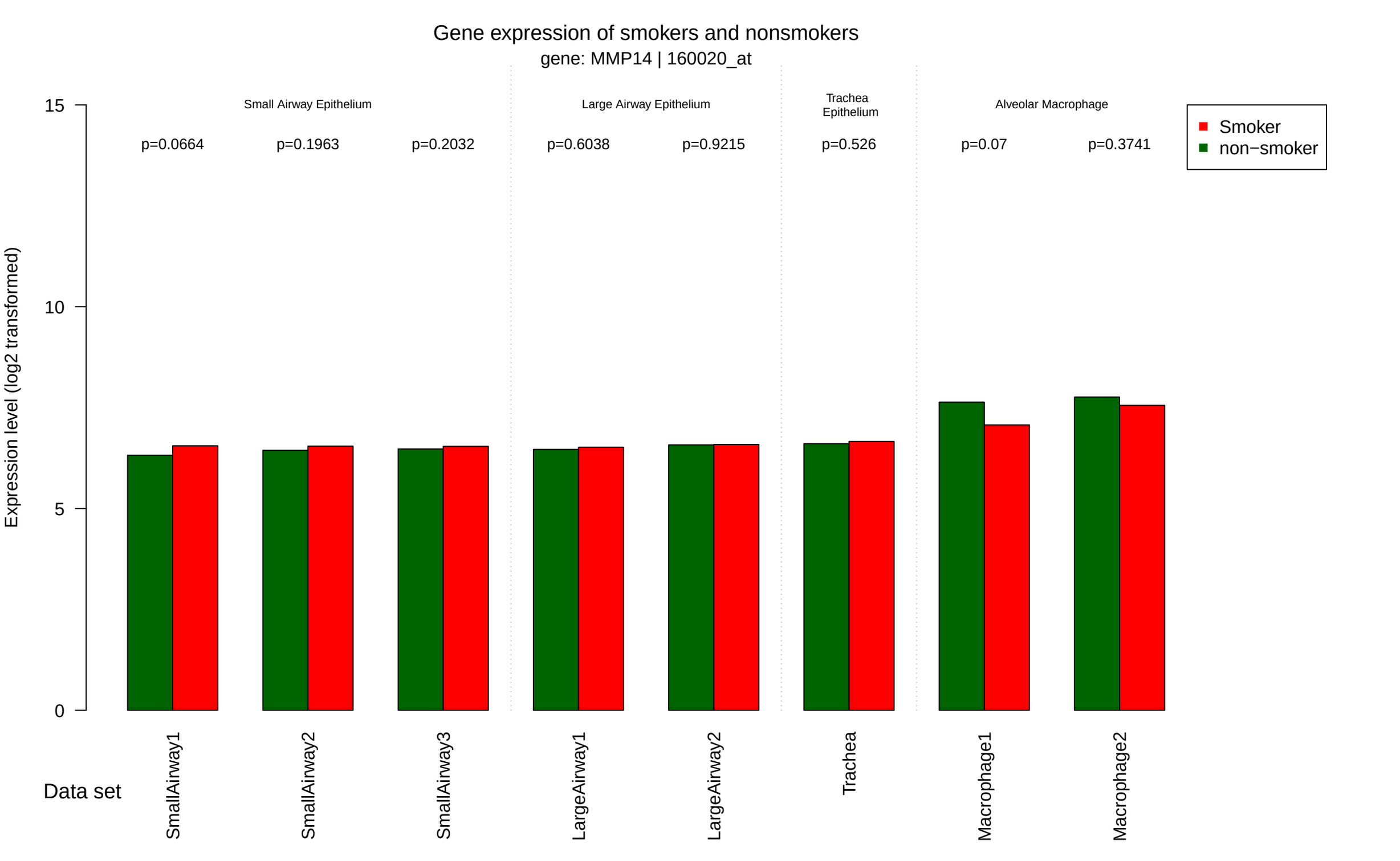
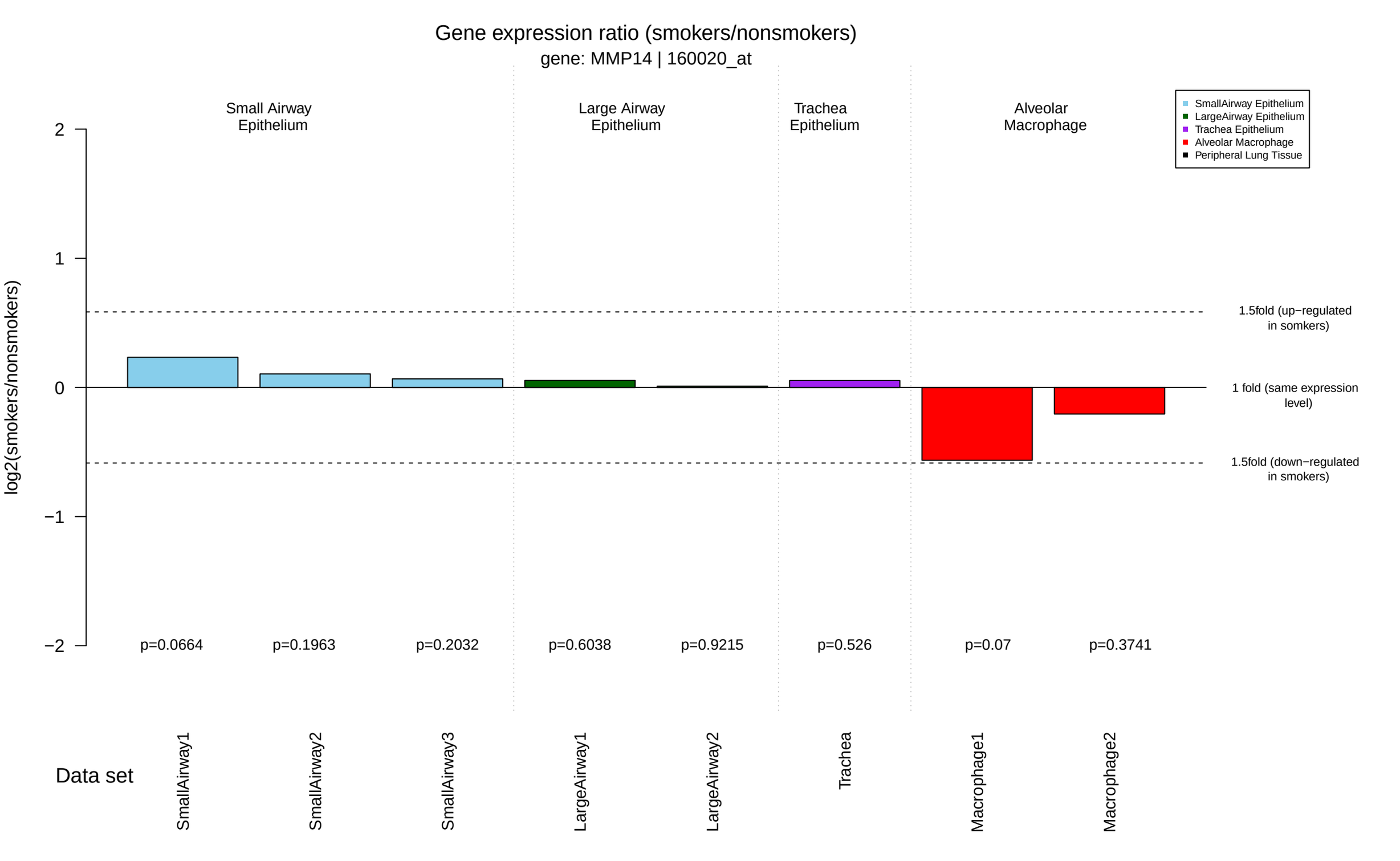
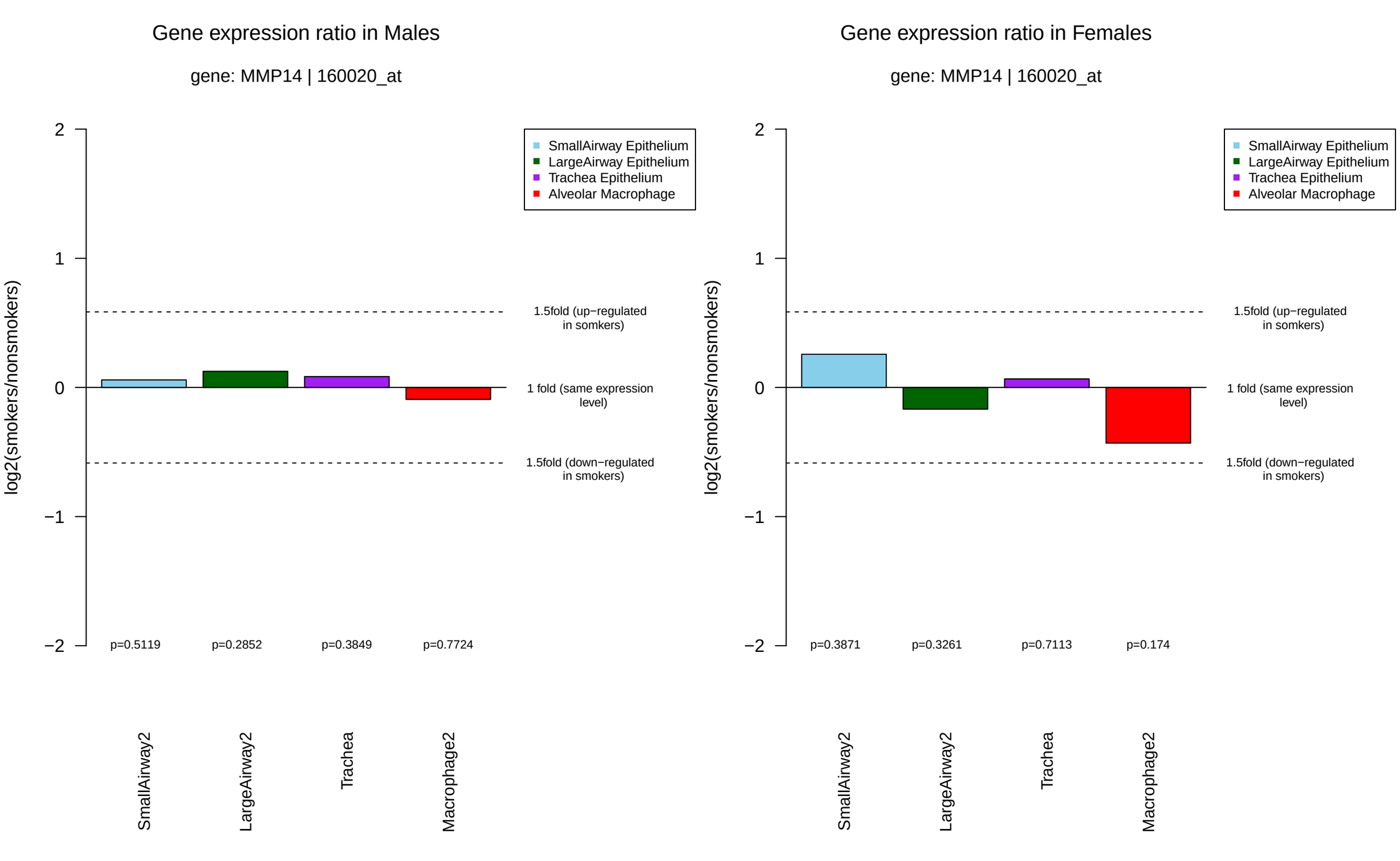
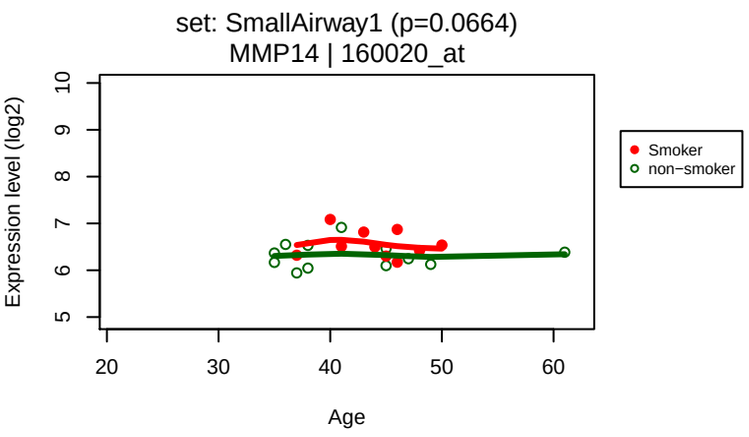
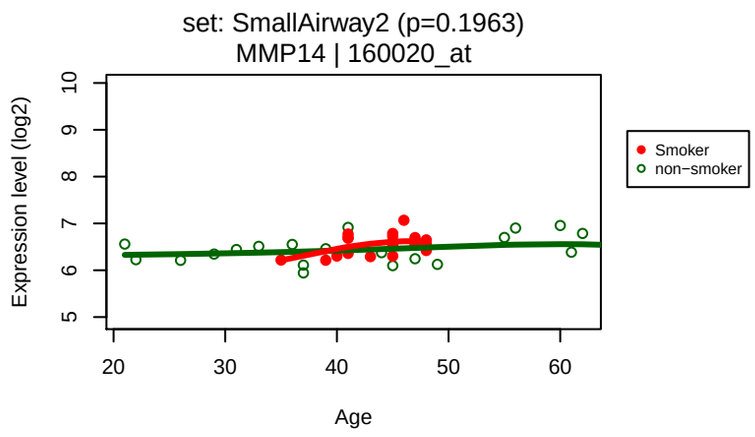
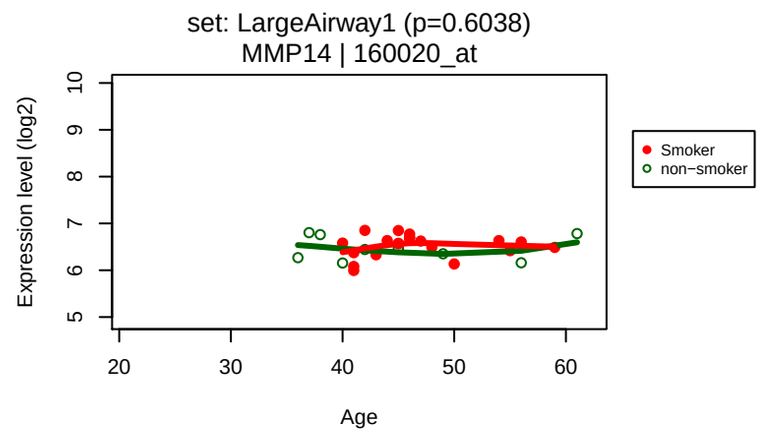

Gene expression of smokers and nonsmokers
gene: MMP14 | 160020_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0664
p=0.1963
p=0.2032
p=0.6038
p=0.9215
p=0.526
p=0.07
p=0.3741
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

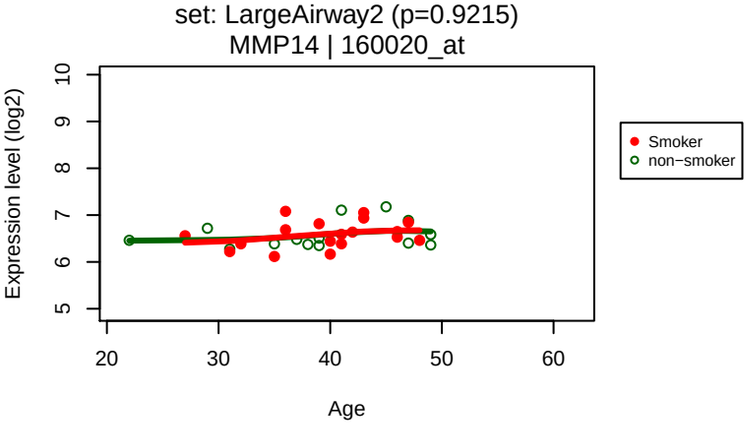

Gene expression ratio (smokers/nonsmokers)
gene: MMP14 | 160020_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0664
p=0.1963
p=0.2032
p=0.6038
p=0.9215
p=0.526
p=0.07
p=0.3741
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

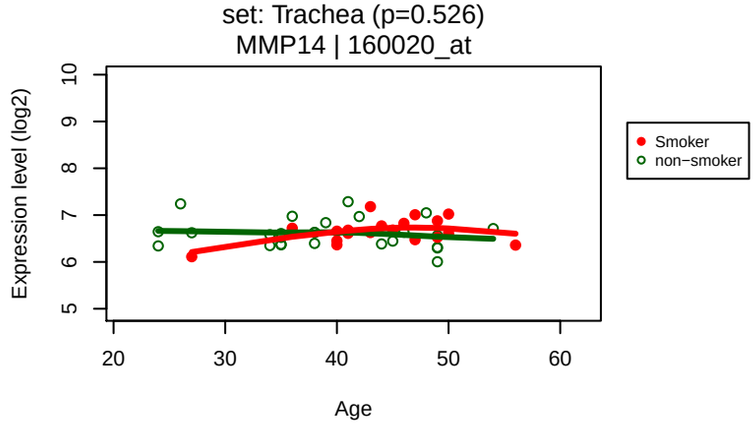

Gene expression ratio in Males
gene: MMP14 | 160020_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5119
p=0.2852
p=0.3849
p=0.7724
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP14 | 160020_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3871
p=0.3261
p=0.7113
p=0.174
SmallAirway2
LargeAirway2
Trachea
Macrophage2

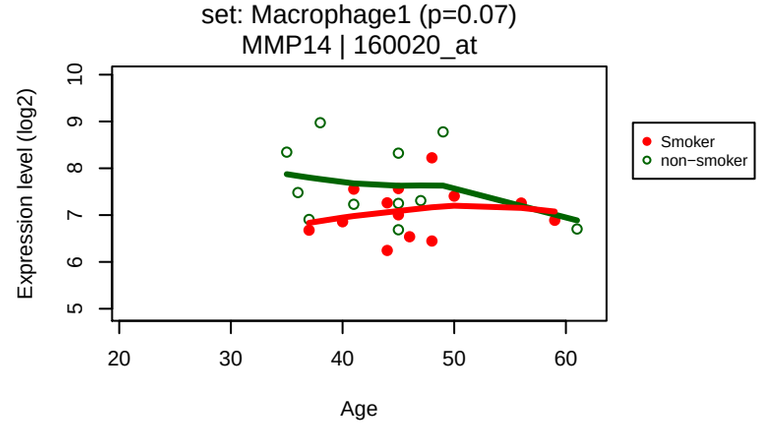

set: SmallAirway1 (p=0.0664)
MMP14 | 160020_at
Expression level (log2)
Age
Smoker
non−smoker

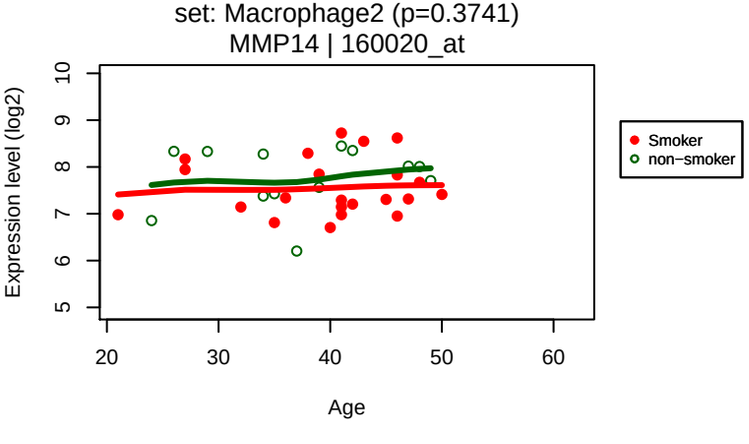

set: SmallAirway2 (p=0.1963)
MMP14 | 160020_at
Expression level (log2)
Age
Smoker
non−smoker

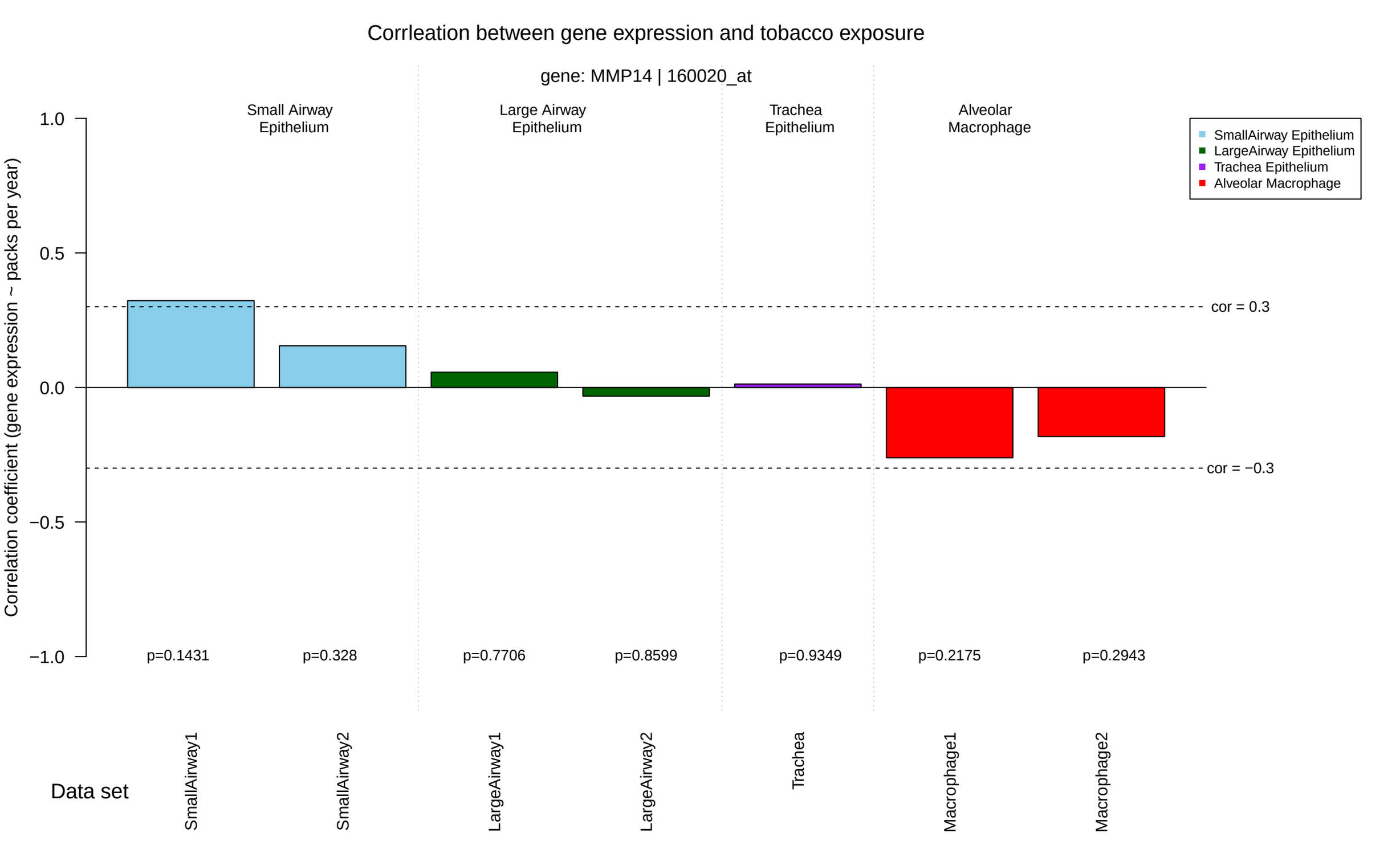

set: LargeAirway1 (p=0.6038)
MMP14 | 160020_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9215)
MMP14 | 160020_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.526)
MMP14 | 160020_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.07)
MMP14 | 160020_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3741)
MMP14 | 160020_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP14 | 160020_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1431
p=0.328
p=0.7706
p=0.8599
p=0.9349
p=0.2175
p=0.2943
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2